

CUSTOMER

PROJECT	x				
MANUFACTURER					
DOCUMENT NO.	x				
DOC TITLE	Ampacity Calculation				
Issue by:	Revision:		Date:		Page: 1 of 3
x					
	Unit	Reference	Clause	Symbol	Formula
					TREFOIL GNDTREFOIL DUCTTREFOIL AIR

DESIGN DATA

Bonding Type	Design data		solid		
Cond Mat	Design data		Al		
Cond size	mm²	Design data	185		
Cond design	Design data		CC		
Cond Dn	mm	Design data	Dc	15.9	15.9
Cond Sc Dn	mm	Design data	Ds	17.6	17.6
Ins Mat	Design data		XLPE		
Ins Dn	mm	Design data	Di	33.9	33.9
Ins Sc Dn	mm	Design data	dx	37.6	37.6
Ins Tn	mm	Design data	Ti	9.19	9.19
Core Dn	mm	Design data	Do	--	--
Lay-up OD	mm	Design data	Da	--	--
Cable Dn	mm	Design data	De	43.4	43.4
Core No	nos	Design data	n 1	1	1
Wire No	nos	Design data	nw	37	37
Cable weight	kg/km	Design data	M	6731	6731
Max cond temp	°C	Design Data	θmax	90	90
Earth fault current	A	Design Data	Ix/1sec 1	1493	1493
Rated voltage	kV	IEC60502-2	Table 1	U	33
Earth voltage	kV	IEC60502-2	Table 1	U0	19
Impulse voltage	kV	IEC60502-2	Table 14	Up	170

PARAMETER SETTING

Soil TR	K.m/W	Input	ρ	1.2	1.2
Multicore factor		Input	fm	1	1
Frequency	Hz	Input	f	50	50
Ambient temp.	°C	Input	θa	30	40
Depth	mm	Input	L	1200	1000
Duct Type		Input		HDPE	HDPE
External diameter of duct	mm	Input	dc	160	110.0
Thickness of duct	mm	Input	Td	0	5.0
Duct temperature	°C	Input	θm	85	85
Core space factor		Input	fs	1	1
Axial Circ distance H	mm	Input	Sh	500	0
Axial Circ distance V	mm	Input	Sv	500	0
Horizontal circuit	nos	Input	nh	1	1
Vertical circuit	nos	Input	nv	1	1
Temp. rise, ambient to max.	°C	IEC60287-1-1 1.4.1.1	Δθ	θmax - θa	60
Core temperature, θc	°C	Iteration	θc		80
Axial Core distance	mm		S		44
Cable space	mm		S'		0
Circuit spacing H	mm		S'h		0
Circuit spacing V	mm		S'v		0
Axial distance cond to cab	mm		c		0
Ins Tn between cond	mm		t		0
Circumscribing radius cond	mm		r1		0.00

PROJECT	x				
MANUFACTURER					
DOCUMENT NO.	x				
DOC TITLE	Ampacity Calculation				
Issue by:	Revision:	Date:	Page:	2 of 3	
x					
	Unit	Reference	Clause	Symbol	Formula
					TREFOIL GNDTREFOIL DUCT TREFOIL AIR

DC RESISTANCE	
---------------	--

AC RESISTANCE	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	
69	
70	
71	
72	
73	
74	
75	
76	
77	
78	
79	
80	
81	
82	
83	
84	
85	
86	
87	
88	
89	
90	
91	
92	
93	
94	
95	
96	
97	
98	
99	
100	

SHEATH LOSS FACTOR	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
14	
15	
16	
17	
18	
19	
20	
21	
22	
23	
24	
25	
26	
27	
28	
29	
30	
31	
32	
33	
34	
35	
36	
37	
38	
39	
40	
41	
42	
43	
44	
45	
46	
47	
48	
49	
50	
51	
52	
53	
54	
55	
56	
57	
58	
59	
60	
61	
62	
63	
64	
65	
66	
67	
68	
69	
70	
71	
72	
73	
74	
75	
76	
77	
78	
79	
80	
81	
82	
83	
84	
85	
86	
87	
88	
89	
90	
91	
92	
93	
94	
95	
96	
97	
98	
99	
100	

CUSTOMER

PROJECT	x						
MANUFACTURER							
DOCUMENT NO.	x						
DOC TITLE	Ampacity Calculation						
Issue by:	Revision:		Date:		Page:		3 of 3
x							
	Unit	Reference	Clause	Symbol	Formula	TREFOIL GND	TREFOIL DUCT
						TREFOIL AIR	

THERMAL RESISTANCE

Thermal Resistant Insulation	K.m/W	IEC60287-2-1		T1	(p/2p)LN[1+(2t/dc)]	0.451	0.451	0.451
Thermal Resistant filler	K.m/W	IEC60287-2-1		T2f		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Bedding	K.m/W	IEC60287-2-1		T2		0.000E+00	0.000E+00	0.000E+00
Thermal Resistant Outercovering	K.m/W	IEC60287-2-1		T3	(p/2p)LN[1+(2t/D)]	0.208	0.130	0.130
Thermal Resistant External	K.m/W	IEC60287-2-1	2.2	T4		2.231	2.377	1.011

CURRENT RATING

Critical Temp Rise of Soil	degC	IEC60287-1-1		dO	Wk(pc/2p)LN(Gf)	0.000E+00	0.000E+00	0.000E+00
Top Equation		IEC60287-1-1	1.4.1	Etop	dO-Wd(0.5T1+n(T2+T3)+T4)-dO	59.9	59.9	50.0
Bottom Equation		IEC60287-1-1	1.4.1	Ebot	RT1+nR(1+lam1)T2+nR(1+lam1+lam2)(T3+T4)	3.438E-04	6.939E-04	3.438E-04
Calculated Amps	A	IEC60287-1-1	1.4.1	I	sqrt(Etop/Ebot)	309	294	381
Calculated MVA	MVA	IEC60287-1-1	1.4.1	MVA	sqrt3*U*I	18	17	22

ELECTRICAL DATA

Inductance	uH/m	IEC60287-1-1	2.3	L	0.2LN(2s/Dm)			
Reactance	uohm/m	IEC60287-1-1	2.3	X	w*Lm*10-3			
Pos Seq Resistance	uohm/m			R1	R'1(1+lam1+lam2)			
Pos Seq Reactance	uohm/m			X1	w*L*10-3			
Pos Seq Impedance	uohm/m			Z1	sqrt(R12+Xm2)			
Zero Seq Resistance	uohm/m			R0	Rx+(n*R'x)			
Capacitance	uF/km	IEC60287-1-1	2.2	C	e/(18LN(Di/ds))	2.119E-10	2.119E-10	2.119E-10
Charging current - Max	A/km	BICC Page 170		Ic max	1.08*w*C*U0			

THERMAL LOSS

Thermal Resistant Total	K.m/W	Mc Graw hill	Page 34	SumT	T1+T2+T3+T4			
Thermal loss	W/m	Mc Graw hill	Page 34	W	(Oc-Oa)/SumT			
Losses - Conductor	W/m	IEC60287-1-1	2.4.2	Wc	3(I2R)	20.18	18.20	30.64
Losses - Dielectric	W/m	IEC60287-1-1	2.2	Wd	3*w*C*U02*tand	0.000E+00	0.000E+00	0.000E+00
Losses - Metal sheath	W/m		IEC60287-1-1	Wm	Wc*lam1			

INDUCED PARAMETER

Normal - Sheath voltage	V/m	IEEE Std 575	Page 29	Us	I*X			
Normal - Circulating current	A	IEEE Std 575	Page 29	Is	sqrt(I2(R/Rs)lam1')			
Transient - Sheath voltage	V/m	IEEE Std 575	Page 29	Ust	Ix*X			
Transient - Circulating current	A	IEEE Std 575	Page 29	Ist	sqrt(Ix2(R/Rs)lam1')	None	None	None

VOLTAGE DROP

Rated voltage	kV	IEC60840	Table 4	U		33	33	33
Pos Seq Impedance	uohm/m			Z1	sqrt(R12+Xm2)			
Calculated Amps	A	IEC60287-1-1	1.4.1	I	sqrt(Etop/Ebot)	309	294	381
Permissible limit	%	Input		a		2.5%	2.5%	2.5%
Permissible voltage drop	V			E	a*U	825	825	825
Voltage drop factor	mV/A/m			R	sqrt3*Z			
Max permissible length	km			L	E/(I*R)			

CRITICAL LENGTH

Critical Length	km			Lmax	sqrt3(I/wCU)x103	None	None	None
-----------------	----	--	--	------	------------------	------	------	------

RECEIVED POWER

Max Feeding Power	MVA	IEC60287-1-1	1.4.1	S				
Max Received Power	MW			P	sqrt[S2-Q2]			
Power Losses	%			Ploss	P/S	%	%	%